Vision-Based Fall Detection Using Object Localization and Motion Analysis

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***Abstract*—This paper introduces a computer vision system for automatic fall detection based on human pose estimation. It states the relevance of the issue, justifying development of a methodology for elderly people and provides set of objectives.**

The comprehensive literature review is exploring the domain of research field. Related works of recent studies are explained to further investigate current trends.

Proposed methodology is based on the transformation of existing dataset using automatic pose estimation by Mediapipe Human pose estimation model. Dataset structure is explained and the set of extracted features is listed. The paper defines both conventional machine learning and deep learning approaches to solve the main issue and compare performances - precision and recall metrics are reported.

The paper provides complete new methodology for live vision based fall detection system, trained on small dataset and achieving performance around 94% of precision.

***Index Terms*—Computer Vision, Human Pose Estimation, automatic annotation, conventional ML, LSTM, TCN, classification.**

I. Introduction

A. Research Problem Statement

Unintentional falls can often result in serious consequences, especially for older individuals, as their coordination declines with age and their muscles become weaker and less controllable.

Quick assistance frequently plays a crucial role that can significantly impact a person's life. However, there are many instances where individuals find themselves alone in a room, hindering others from providing emergency help [7]. Fortunately, nowadays, CCTV cameras are quite common, and if they include the right technologies, a quick response to a fall can be implemented. This paper aims to create a computer vision system that can accurately detect when individuals fall and distinguish fall action with non-fall actions in a very precise way.

B. Research Objectives

This study focuses on set of following objectives:

- Suggest practically applicable methodology for fall detection.
- Use deep-learning based object (human) localization.
- Use skeleton-based motion analysis.
- Performance comparison with previous work.

II. Literature Review

Rougier et al [6] proposed a fall detection method based on analyzing the deformation of the human

figure in a video sequence using Shape Context Matching and Procrustean distance analysis. The method is trained on normal (no-fall) actions and classifies falls using a Gaussian Mixture Model (GMM), which pays attention on for abrupt silhouette shape changes and subsequent immobility. Testing is conducted using realistic video data with occlusions and complex conditions resulted in accuracy about 97% using a multi-camera system, which performs much better than traditional velocity analysis methods or accelerometers.

Authors from the study [5] proposed a deep learning based fall detection method by combining 3D CNN to extract spatio- temporal features and LSTM with visual attention mechanism to extract key regions in the frame. Unlike traditional methods, this hybrid model does not require a large dataset of falls, but is trained on the Sports-1M sports dataset containing a variety of movements. A 3D CNN is then used to extract kinematic features from the video of falls, and LSTM refines the regions of attention. The method has shown 100% accuracy on Multiple Cameras Fall Dataset and performs better than other approaches for fall classification.

Anishchenko [2] used transfer learning, adapting AlexNet CNN for fall detection based on CCTV data. Training was performed on a real dataset which contains variety of lighting conditions, different viewing angles, and inconsistent surfaces. The method achieved Cohen's kappa metrcis of 0.93 for known conditions and 0.60 for new conditions, which shows a decrease in accuracy with environmental changes. The authors note the need to increase the training dataset and utilize temporal relationships between frames to improve classification reliability.

Yu et al. [8] proposed a method for fall detection based on a deep convolutional neural network (CNN) that analyzes video sequences without the need for wearable sensors. The method uses background subtraction to extract a person's silhouette and further classify the pose using CNN. Four states are recognized: standing, sitting, reclining, and lying, after which the system analyzes the position of the body relative to the floor. If a lying posture is detected in the floor region, a fall is recorded. The CNN uses rectangular bounding boxes of 30×30 pixels as input data. The testing on 3,216 images with 10 subjects showed activity classification accuracy of 97.83%, and testing on 98 video scenarios with falls performed 99% successful detections with a very low false positive rate (1%). The main errors are related to the overlap between reclining and lying poses, which the authors propose to correct by using multi-camera systems or adding audio data.

Zhang et al. [9] proposed three various computer vision fall detection approaches for monitoring elderly people at home. The first method analyzes centroid position of the human body in a single frame and detects a fall if the centroid falls below some fixed threshold. The second method uses a sequence of frames and determines a fall by a set of calculated features: the number of frames of the fall, the change in body position (in pixels) and the maximum speed of the fall. The third method utilizes statistical analysis and uses probability models instead of thresholds to recognize falls more accurately but less quick. The authors apply cross-validation and model fall probabilities using Gaussian and nonparametric distributions. Experiments on 80,250 frames show that the statistical method has the highest accuracy, capturing all 8 falls while minimizing false positives. The main limitation of the work is the small amount of training data, which the authors plan to increase by integrating multi-camera systems.

Alam et al. [1] developed a method for fall detection based on deep learning and computer vision by extracting spatio-temporal features from video sequences. Spatial features include silhouette segmentation and body shape change analysis, bounding box aspect ratio and centroid displacement. Temporal features are calculated based on optical flow, pose trajectories, fall velocity, and subsequent period of immobility. In some cases, multimodal features such as depth camera or accelerometer data are integrated to boost accuracy in challenging lighting conditions. The model uses convolutional neural networks (CNNs) to extract spatial features and recurrent networks (LSTMs) to analyze temporal dependencies, allowing to distinguish falls from normal movements (e.g.,rapid squatting). Experi-

mental tests on public datasets showed high fall classification accuracy (about 98%), with the main errors remaining false positives in cases of sudden movements which often are very similar to falls.

III. Methodology

A. Dataset

To solve the problem, it was decided to use the Fall Detection dataset (available for free use). The dataset contains videos shot in realistic surveillance setting [4]. Each video size is 320 by 240 pixels, at frame rate of 25 fps.

It consists of fall and no-fall directories, which also includes annotation files: bounding box coordinates for each frame. However, for our approach, some changes were made to dataset which will be discussed later on this section.

B. Proposed Approach

1) Dataset Transformation: The approach idea is to use pose estimation technology to reconstruct the dataset. We will use MediaPipe model to extract human pose coordinates for video annotation.

Then, each annotation file undergoes consistent feature extraction. Those features are:

- Mean center (x-axis)
- Mean center (y-axis)
- Standard deviation (x-axis)
- Standard deviation (y-axis)
- Mean aspect ration
- Maximum velocity center
- Mean acceleration center
- Total horizontal displacement
- Total vertical displacement

As an additional feature, we considered missing data as well. Since there is a fact of low confidence predictions for creating an annotation file, fast position changes just in case of fall action generate large amount of missing values, so the additional feature to be added is the **missing data ratio.**

The transformation is conducted by altering the annotation file schema. he initial structure is presented on Figure 1. It contains starting and ending frame of reported fall action. First column represents frame number, second column provides the

$$\text{Annotation File} = \begin{bmatrix} \text{Start Frame} & & & & & \\ \text{End Frame} & & & & & \\ \text{Frame} & \text{ID} & x_1 & y_1 & x_2 & y_2 \\ 1 & 1 & 35 & 15 & 106 & 140 \\ 2 & 1 & 32 & 13 & 102 & 138 \\ 3 & 1 & 32 & 14 & 106 & 138 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ n & 1 & x_1^{(n)} & y_1^{(n)} & x_2^{(n)} & y_2^{(n)} \end{bmatrix}$$

Fig. 1. Initial Dataset Annotation Schema

$$\text{Transformed Annotation File} = \begin{bmatrix} \text{Start Frame} & & & & & & & \\ \text{End Frame} & & & & & & & \\ \text{Frame} & x_1 & y_1 & x_2 & y_2 & \cdots & x_{20} & y_{20} \\ 0 & 0.738 & 0.401 & 0.730 & 0.552 & \cdots & 0.684 & 0.721 \\ 1 & 0.738 & 0.400 & 0.733 & 0.548 & \cdots & 0.694 & 0.700 \\ 2 & -1 & -1 & -1 & -1 & \cdots & -1 & -1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ n & x_1^{(n)} & y_1^{(n)} & x_2^{(n)} & y_2^{(n)} & \cdots & x_{20}^{(n)} & y_{20}^{(n)} \end{bmatrix}$$

Fig. 2. Transformed Dataset Annotation Schema

ID of a person and each consecutive column is a coordinate value for bounding box annotation.

After transformation, the structure of the annotation file is slightly changed, which is visible on Figure 2. The main difference is larger number of descriptive coordinates, there are twenty coordinate pairs captured from lower body keypoints using Mediapipe Human Pose estimation model.

2) General Approach: The system itself requires well structured and balanced dataset, but in isolation it is just a binary classification case. General approach algorithm is presented on Figure 3. As it was already stated, prediction model will be trained on transformed dataset. However, exact fall detection system will include human estimation block to capture human body position from the video and make a prediction using machine learning approach.

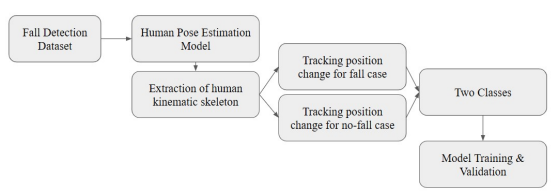

Fig. 3. General Approach Schema

3) Prediction Models: Dataset transformation provides comprehensive set of features for both desired classes: fall and no-fall actions. Several models will be trained and analyzed.

For prediction, both conventional machine learning and deep learning methods were utilized. For machine learning approaches, following classifiers were trained:

- Support Vector classifier
- XGBoost
- Random Forest
- Logistic Regression
- K-Nearest Neighbors
- Gradient Boosting
- Naive Bayes

The evaluation of each model was conducted with focus on predicting positive class: usage of such metrics as precision and recall is justified. Confusion matrix for this case is presented in table 1.

Precision is calculated as: $\frac{TP}{TP+FP}$, and Recall: $\frac{TP}{TP+FN}$.

Those metrics allow to evaluate how good model performs on predicting positive class.

TABLE I
CONFUSION MATRIX FOR FALL DETECTION (BINARY CLASSIFICATION)

		Predicted	
		No-Fall (0)	**Fall (1)**
Actual	**No-Fall (0)**	TN	FP
	Fall (1)	FN	TP

Each model is trained using 80:20 split. All features are scaled using Standard Scaler. The output of each model prediction is a pair of probabilities, describing the confidence of no-fall and fall actions.

In addition, deep learning approaches were used to compare performances of neural networks for that task. Same methodology and data were used to train deep learning models with slightly different pre-processing. Following models were tested:

- Long-Short Term Memory (LSTM)
- Temporal Convolutional Networks (TCN)

For deep learning models evaluation, same metrics were calculated: precision and recall.

IV. RESULTS AND DISCUSSION

A. Machine Learning Approach

In this section, we will outline the results we have achieved. They will be compared with the results of Charifi et al. [3] which served as the baseline for our study. The main metrics for comparison will be Precision and Recall.

1) Comparison with Baseline: The method of Charifi et al. [3] achieved precision 94.23% and recall 98.03% using STHF for automatic annotation and SVM as a classifier. Our system using skeleton coordinates as annotation with SVM managed to achieve precision = 92% and recall = 92% as it can be seen from Table 2. Although our results do not exceed the baseline, using the skeleton approach helps to create easily interpretable features. Since they clearly display human movement and behavior, in our study there is no need for complex filtering and transformation of the obtained data to create the corresponding features. Due to this, using the Random Forest classifier, we were able to achieve results of precision 98.00%, which outperforms the baseline model. The final recall, on the contrary, turned out to be worse than that of the original study author and equals 94.00%. This indicator shows that our model predicts more False Positives, but this is not a problem. If a person is in good condition, and the model predicts a fall, this is not so critical compared to a situation when a person needs help, and the model did not detect it. Random Forest proved to be the most accurate, because it works well with noisy data, as well as in moments when data may be partially missing, which is observed in our case. The structure of this classifier also helps to detect nonlinear dependencies, as well as reduce overfitting. In comparison, due to their simplicity the Logistic Regression and KNN models demonstrated the worst results.

2) Feature Importance: After analyzing the importance of features, we found that the standard deviation and the mean value of the vertical center coordinate (std_center_y and mean_center_y) had the greatest impact on the classification. In fact, this is not surprising, because it is a strong vertical displacement of the body that most strongly signals a fall. Other important features are the x-shift and the

TABLE II
COMPARISON OF PRECISION AND RECALL WITH CHARFI ET AL.'S AUTOMATICALLY ANNOTATED MODEL

Model	Precision (%)	Recall (%)
Charfi et al. [3]	94.23	98.03
Our SVC-based Model	92.00	92.00
XGBoost	93.00	90.00
Random Forest	**98.00**	**94.00**
Logistic Regression	86.00	85.00
K-Nearest Neighbors	85.00	82.00
Gradient Boosting	95.00	89.00
Naive Bayes	85.00	90.00

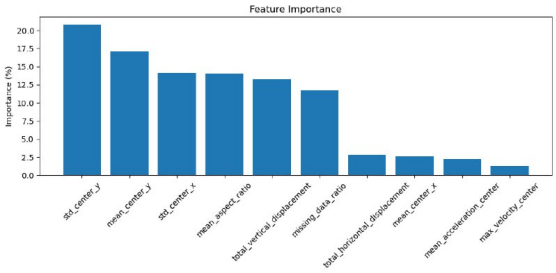

Fig. 4. Feature Importance Diagram

mean_aspect_ration, which show the displacement of body proportions and the horizontal change. This is also quite reasonable, because when falling, the human body often deforms and instinctively compresses to cushion and soften the fall. Less significant were the maximum speed of the center and the average acceleration. Presumably, this is due to the data itself. Since people in the dataset simulate falls, even despite the mattress placed under the place of the fall to prevent injury, people subconsciously tried to fall smoother, which affected the classifier's recording of smooth falls. What really surprised us was that our theory was confirmed. Using the missing data ratio feature affects the result by as much as 12%. This is because when falling and deforming the body, it is difficult for Mediapipe to record the position of the body and determine the points of the skeleton. Therefore, when a gap in the data appeared, it became clear that it was caused by a sharp change in the body in space, which signals a fall.

Thus, our approach demonstrates comparable performance with the baseline approach by Charfi et al. [3] At the same time, we managed to achieve several advantages, such as a simplified feature architecture and their easier interoperability.

B. Deep Learning Approach

In this work we have conducted several experiments with Deep Learning models. We used the same metrics for comparison: precision and recall.

1) Comparison with traditional Machine Learning models: We have used the same feature extraction methods. However, the pre-processing technique is different. Instead of summarizing each feature for the entire video, each frame is extracted with the feature vector in order to capture temporary patterns.

Our LSTM model achieved precision = 71% and recall = 97%. The temporal convolution network(TCN) model achieved precision = 92% and recall = 86%. Although TCN achieved decent results, neither of them outperformed the baseline mode nor some of the other models tested in this paper.

2) Possible reasons: Most of the traditional Machine Learning algorithms outperformed Deep Learning models.

We have several possible reasons which can explain this result.

First, the size of the data set is insufficient. DL models usually require a lot of parameters to generalize well.

Secondly, the features are already informative enough. The feature importance diagram (Fig. 4) demonstrates that most important features are related to the change of the human body on the y-axis. These are themselves meaningful features.

Third, DL models tend to overcomplicate tasks and overfit. The limited data set exacerbates these problems.

Thus, conventional ML models are sufficient for fall prediction task.

V. CONCLUSION AND FUTURE WORK

A. Conclusion

This paper presents a computer vision system for human fall detection.

TABLE III
PRECISION AND RECALL FOR LSTM AND TCN MODELS

Model	Precision (%)	Recall (%)
LSTM	71	97
TCN	92	86

The system used MediaPipe to extract key features for human pose estimation. The system was implemented using a wide range of conventional Machine Learning methods, as well as Deep Learning methods. The Random Forest model outperformed the baseline model with a precision score of 98% and a recall score of 94%.

Deep Learning models demonstrated decent results. However, they were unable to outperform traditional Machine Learning methods due to the limited data set size and sufficient informativeness of the existing features.

B. Future Work

For the future work, we are planning:

- Model deployment.
- Capturing time point (or frame range) where fall did happen.
- Instead of video input, try streaming input.

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